Example1: Dataset:

Person	Age	PSA	MRI	Class
1	85	5.1	3	Cancer
2	54	4.3	2	Healthy
3	50	1.2	1	Healthy
4	65	4.9	3	Cancer
5	67	5.6	4	Cancer
6	56	6.7	5	Cancer
7	63	6.5	2	Healthy
8	70	3.0	3	Healthy
9	50	4.9	3	Healthy

Suppose new patient (Test Data)

Age = 68, PSA = 6.0, MRI = 3: Prediction=???? (Cancer or Healthy)

Solution:

- Features **(M)**= **Age, PSA, MRI**
- Target = **Class**
- Number of individuals **(N)= 9**

Gini Formula

$$Gini = 1 - (p_1^2 + p_2^2 + \ldots + p_n^2)$$

Where

- p_i= probability of each class.

Classes here: **Cancer, Healthy**

Step 1: Count Class Distribution and Parent Node Gini

From the dataset:

Cancer = **4**
Healthy = **5**
Total = **9**

$$p(Cancer) = 4/9 = 0.44$$
$$p(Healthy) = 5/9 = 0.56$$

Parent Node Gini

$$Gini = 1 - (0.44^2 + 0.56^2)$$
$$= 1 - (0.1936 + 0.3136)$$
$$= 1 - 0.5072$$
$$Gini = 0.4928$$

Parent node impurity ≈ **0.49**

Step 2: Tree 1 (Feature = PSA)

Split:

PSA ≤ 5
PSA > 5

Group 1 (PSA ≤ 5)

Persons = 2,3,4,8,9

Healthy = 4
Cancer = 1

$$Gini = 1 - (0.8^2 + 0.2^2)$$
$$= 1 - (0.64 + 0.04)$$
$$= 0.32$$

Group 2 (PSA > 5)

Persons = 1,5,6,7

Cancer = 3
Healthy = 1

$$Gini = 1 - (0.75^2 + 0.25^2)$$
$$= 1 - (0.5625 + 0.0625)$$
$$= 0.375$$

Weighted Gini

$$(5/9)(0.32) + (4/9)(0.375)$$
$$= 0.178 + 0.167$$
$$= 0.345$$

Tree 1 split **improves impurity**.

Tree:

PSA

/ \

≤5 >5

Healthy Cancer

Step 3: Tree 2 (Feature = MRI)

Split:

MRI ≤ 2
MRI > 2

Group 1

Persons: 2,3,7

Healthy = 3
Cancer = 0

$$Gini = 1 - (1^2 + 0^2)$$
$$= 0$$

Pure node.

Group 2

Persons: 1,4,5,6,8,9

Cancer = 4
Healthy = 2

$$p(Cancer) = 4/6$$
$$p(Healthy) = 2/6$$
$$Gini = 1 - (0.67^2 + 0.33^2)$$
$$= 1 - (0.4489 + 0.1089)$$
$$\approx 0.442$$

Weighted Gini

$$(3/9)(0) + (6/9)(0.442)$$
$$= 0.295$$

Tree 2:

```
     MRI
    /  \
   ≤2    >2
 Healthy  Cancer
```

Step 4: Tree 3 (Feature = Age)

Split:

Age ≤ 60
Age > 60

Group 1

Persons: 2,3,6,9

Healthy = 3
Cancer = 1

$$Gini = 1 - (0.75^2 + 0.25^2)$$
$$= 1 - (0.5625 + 0.0625)$$
$$= 0.375$$

Group 2

Persons: 1,4,5,7,8

Cancer = 3
Healthy = 2

$$Gini = 1 - (0.6^2 + 0.4^2)$$
$$= 1 - (0.36 + 0.16)$$
$$= 0.48$$

Weighted Gini

$$(4/9)(0.375) + (5/9)(0.48)$$
$$= 0.166 + 0.267$$
$$= 0.433$$

Tree 3:

```
     Age
    /  \
  ≤60   >60
 Healthy  Cancer
```

Step 5: Random Forest (3 Trees)

```
            RANDOM  FOREST
                  |
     ┌────────────┼────────────┐
     |            |            |
  Tree1        Tree2        Tree3
   PSA          MRI          Age
```

Step 5: Majority Voting Example

Suppose new patient:

Age = 68
PSA = 6.0
MRI = 3

Predictions:

Tree 1 → Cancer
Tree 2 → Cancer
Tree 3 → Cancer

Votes:

Cancer = 3
Healthy = 0

Final Prediction:

Cancer

Example2: Data Set

Person	Income (Lakh)	Credit Score	Age	Loan
1	3	650	25	Rejected
2	4	700	30	Approved
3	2	600	22	Rejected
4	6	720	40	Approved
5	7	750	45	Approved
6	3	680	28	Rejected
7	5	710	35	Approved
8	4	690	32	Approved

Solution:

Total **N = 8**

Approved = **5**
Rejected = **3**

Step 1: Parent Node Gini

$$Gini = 1 - (P_1^2 + P_2^2)$$
$$P_{Approved} = 5/8$$
$$P_{Rejected} = 3/8$$
$$Gini = 1 - ((5/8)^2 + (3/8)^2)$$

$$= 1 - (0.39 + 0.14)$$
$$= 0.47$$

Parent impurity = **0.47**

Step 2: Tree 1 (Income Split)

Split:

Income ≤ 4
Income > 4

Group 1 (≤4)

Persons: 1,2,3,6,8

Approved = 2
Rejected = 3

$$Gini = 1 - (0.4^2 + 0.6^2) = 0.48$$

Group 2 (>4)

Persons: 4,5,7

Approved = 3
Rejected = 0

$$Gini = 0$$

Weighted Gini

$$(5/8)(0.48) + (3/8)(0)$$
$$= 0.30$$

Tree 1

```
    Income
     /        \
≤4            >4
Rejected    Approved
```

Step 3: Tree 2 (Credit Score Split)

Split:

Credit ≤ 690
Credit > 690

Group 1

Persons: 1,3,6,8

Approved = 1
Rejected = 3

$$Gini = 1 - (0.25^2 + 0.75^2) = 0.37$$

Group 2

Persons: 2,4,5,7

Approved = 4
Rejected = 0

$$Gini = 0$$

Weighted Gini

$$= 0.18$$

Tree 2

Credit Score

/ \

≤690 >690

Rejected Approved

Step 4: Tree 3 (Age Split)

Split:

Age ≤ 30
Age > 30

Group 1

Persons: 1,2,3,6

Approved = 1
Rejected = 3

Gini = **0.37**

Group 2

Persons: 4,5,7,8

Approved = 4
Rejected = 0

Gini = **0**

Tree 3

Age

/ \

≤30 >30

Rejected Approved

Step 5: Bootstrap Samples

Assume the following samples were used for training the trees.

Tree	Training Samples	OOB Samples
Tree1	1,2,3,4,5,6,7	8
Tree2	1,3,4,5,6,7,8	2
Tree3	2,3,4,5,6,7,8	1

Step 5: OOB Predictions

Person 8 (Tree1 OOB)

Income = 4 → ≤4

Prediction = **Rejected**

Actual = **Approved**

❌ Incorrect

Person 2 (Tree2 OOB)

Credit = 700 → >690

Prediction = **Approved**

Actual = **Approved**

✓ Correct

Person 1 (Tree3 OOB)

Age = 25 → ≤30

Prediction = **Rejected**

Actual = **Rejected**

✓ Correct

Step 6: OOB Error

Total OOB predictions = **3**

Incorrect predictions = **1**

$$OOB\ Error = \frac{1}{3}$$
$$OOB\ Error = 0.33$$

Final Result

Metric	**Value**
Total OOB predictions	3
Incorrect predictions	1
OOB Error	**0.33**
OOB Accuracy	**67%**

Final Random Forest Prediction Example

New Applicant:

Income = **5**
Credit Score = **705**
Age = **33**

Tree 1 → Approved
Tree 2 → Approved
Tree 3 → Approved

Majority Vote:

Approved = **3**

Final Prediction = **Loan Approved**